# Data Communication & Computer Networks — Topic 4: Networking Devices

**Practice Test Booklet** — 53 MCQs (2-column, without solutions) · Easy + Medium + Hard mix · Har question ke saath **Source** · Print ke liye · Reference: Forouzan + India-level & worldwide exam papers (GATE/NET/KVS/EMRS/DSSSB/SSC/RRB/IBPS) · Prepared for: RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)

## Section A — Physical-Layer Devices (Hub, Repeater, Modem & Simple Devices)

**Q1. Which device operates only at the Physical layer (Layer 1) of the OSI model?**

OSI model की केवल Physical layer (Layer 1) पर कौन-सा device काम करता है?

**(A)** Switch **(B)** Hub
**(C)** Router **(D)** Bridge

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch) · also CN Topic Test-6] · Easy

**Q2. Which of the following pairs lists the network devices used at the physical layer?**

निम्न में से कौन-सी जोड़ी physical layer पर उपयोग होने वाले network devices बताती है?

**(A)** Gateway, Bridge **(B)** Router, Repeater
**(C)** Hub, Switch **(D)** Repeater, Hub

[Source: Random Test- Rajasthan PYQ (OCR), p12 — Q93 (options reconstructed from screenshot)] · Medium

**Q3. Which of the following are network connecting devices? 1. Repeater 2. Bridge 3. Hub 4. Bluetooth**

निम्न में से कौन network connecting devices हैं? 1. Repeater 2. Bridge 3. Hub 4. Bluetooth

**(A)** 1, 2 and 3 **(B)** 1, 2 and 4
**(C)** 2, 3 and 4 **(D)** 1, 3 and 4

[Source: Unit Test -2 Solutions (OCR), p11] · Medium

**Q4. Which of the following statements are TRUE? 1. Switch reduces collision domain. 2. Hub creates a separate collision domain for each port. 3. Router separates broadcast domains.**

निम्न में से कौन-से कथन सही हैं? 1. Switch collision domain को कम करता है। 2. Hub हर port के लिए अलग collision domain बनाता है। 3. Router broadcast domains को अलग करता है।

**(A)** 1 and 2 only **(B)** 2 and 3 only
**(C)** 1 and 3 only **(D)** 1, 2 and 3

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q26] · Medium

**Q5. Which device combines transmission from various I/O devices into one line?**

कौन-सा device विभिन्न I/O devices के transmission को एक line में मिलाता है?

**(A)** Modifier **(B)** Multiplexer
**(C)** Concentrator **(D)** Gateway

[Source: Unit Test-4 Solution (OCR), p22] · Medium

**Q6. A file server is a _____ which is used to store files received from various client machines.**

File server एक _____ है जिसका उपयोग विभिन्न client machines से प्राप्त files संग्रहित करने के लिए होता है।

**(A)** Specialized computer system **(B)** Modem
**(C)** Router **(D)** Transponder

[Source: Unit Test-4 Solution (OCR), p24] · Medium

**Q7. In a data-entry office, computer centres on the 1st and 4th floors are connected by LANs, but signals take time to travel. Which hardware can improve the data transmission?**

एक data-entry office में 1वीं और 4वीं मंज़िल के computer centres LAN से जुड़े हैं, पर signals में समय लगता है. बेहतर transmission के लिए कौन-सा hardware लगाएँ?

**(A)** Repeater **(B)** Router
**(C)** Computer **(D)** Modem

[Source: RSMSSB Informatics Assistant 2023 (via examsmantra solved paper)] · Medium

**Q8. A device that regenerates a weak signal bit-by-bit to its original strength is a:**

कमज़ोर signal को bit-by-bit पुनः मूल शक्ति तक लाने वाला उपकरण है:

**(A)** Repeater **(B)** Amplifier
**(C)** Bridge **(D)** Router

[Source: Practice — original (textbook-based)] · Medium

**Q9. Which of the following devices operates only in the physical layer?**

निम्न में से कौन-सा device केवल physical layer पर काम करता है?

**(A)** Two-layer switch **(B)** Router
**(C)** Repeater **(D)** Bridge

[Source: Kerala PSC repeated PYQs (via pscarivukal)] · Hard

**Q10. A device that operates at the physical layer of the OSI model is:**

OSI model की physical layer पर काम करने वाला device है:

**(A)** Bridge **(B)** Router
**(C)** Repeater **(D)** All of these

[Source: SBI/IBPS Computer Awareness Capsule 2018 (IBT India)] · Hard

**Q11. All ports of a hub belong to how many collision domains?**

Hub के सभी ports कितने collision domains में आते हैं?

**(A)** One / एक **(B)** Equal to number of ports / Ports की संख्या के बराबर
**(C)** Two / दो **(D)** Zero / शून्य

[Source: Practice — original (textbook-based)] · Hard

**Q12. A 24-port hub creates how many collision domains?**

24-port hub कितने collision domains बनाता है?

**(A)** 1 **(B)** 12
**(C)** 24 **(D)** 48

[Source: Practice — original (textbook-based)] · Hard

## Section B — Switch / NIC / Bridge

**Q13. Which device stores MAC addresses in a CAM table?**

कौन-सा device MAC addresses को CAM table में संग्रहित करता है?

**(A)** Hub **(B)** Router
**(C)** Switch **(D)** Modem

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

**Q14. A router is connected between two switches. If a broadcast frame is sent on Switch-A, what will Switch-B receive?**

दो switches के बीच एक router जुड़ा है। यदि Switch-A पर broadcast frame भेजा जाए, तो Switch-B क्या प्राप्त करेगा?

**(A)** The broadcast **(B)** A unicast copy
**(C)** Nothing **(D)** A multicast

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

**Q15. What does a MAC address represent?**

MAC address किसका प्रतिनिधित्व करता है?

**(A)** The location of the device in the network **(B)** The unique identity of the NIC
**(C)** Internet connectivity **(D)** The routing path

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

**Q16. Which device reduces collision domains (each port has its own collision domain) but does NOT break broadcast domains?**

कौन-सा device collision domains को कम करता है (हर port का अपना collision domain) लेकिन broadcast domains को तोड़ता नहीं?

**(A)** Switch **(B)** Router
**(C)** Hub **(D)** Gateway

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test-6] · Medium

**Q17. A switch receives a frame whose destination MAC is NOT in its MAC table. What does the switch do?**

एक switch को frame मिलता है जिसका destination MAC उसके MAC table में नहीं है। Switch क्या करेगा?

**(A)** Send the frame to the router only **(B)** Flood the frame out of all its ports (same broadcast domain)
**(C)** Drop the frame **(D)** Broadcast it on the entire internet

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

**Q18. A switch has 4 VLANs configured. How many broadcast domains exist?**

एक switch पर 4 VLANs configure हैं। कितने broadcast domains होंगे?

**(A)** 1 **(B)** 2
**(C)** 4 **(D)** Depends on the router

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

**Q19. In a router-on-a-stick setup with VLANs, which device handles VLAN tagging and which handles inter-VLAN routing?**

VLANs वाले router-on-a-stick setup में VLAN tagging कौन करता है और inter-VLAN routing कौन?

**(A)** Switch – VLAN tags, Router – routing **(B)** Router – VLAN tags, Switch – routing
**(C)** Both switch and router do VLAN tagging **(D)** Neither does VLAN tagging

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

**Q20. A packet travels: wired PC → Switch → Access Point → Wireless laptop. Which devices operate at Layer 2 only?**

एक packet यात्रा करता है: wired PC → Switch → Access Point → Wireless laptop। कौन-से devices केवल Layer 2 पर काम करते हैं?

**(A)** Switch only **(B)** Access Point only
**(C)** Both switch and access point **(D)** Neither

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

**Q21. Which of the following component is responsible for network communication in a computer system?**

कंप्यूटर सिस्टम में network communication के लिए कौन-सा component ज़िम्मेदार होता है?

**(A)** UPS **(B)** Ethernet card (NIC)
**(C)** SVGA card **(D)** SMPS

[Source: Unit Test-4 Solution.pdf (OCR), p18] · Medium

**Q22. LANs can be connected by which of the following devices that operate on the data link layer?**

LANs को data link layer पर काम करने वाले निम्न में से किस device से जोड़ा जा सकता है?

**(A)** Hub **(B)** Bridges
**(C)** HDLC **(D)** Tunnel

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Medium

**Q23. A switch learns the port for a MAC address by:**

Switch किसी MAC address का port कैसे सीखता है?

**(A)** Reading the source MAC of incoming frames / आने वाले frames के source MAC से **(B)** Asking the router / Router से पूछकर
**(C)** Reading the IP address / IP address पढ़कर **(D)** Broadcasting every frame / हर frame broadcast करके

[Source: Practice — original (textbook-based)] · Medium

**Q24. VLANs on a switch are used to create:**

Switch पर VLAN बनाए जाते हैं:

**(A)** More collision domains / अधिक collision domains **(B)** Separate broadcast domains / अलग-अलग broadcast domains
**(C)** More repeaters / अधिक repeaters **(D)** Faster cables / तेज़ cables

[Source: Practice — original (textbook-based)] · Medium

**Q25. Connecting each switch port to a single host so that every port becomes its own collision domain is called:**

Switch के हर port पर एक host लगाकर उसे अलग collision domain बनाना कहलाता है:

**(A)** Microsegmentation **(B)** Trunking
**(C)** VLAN hopping **(D)** Bridging loop

[Source: Practice — original (textbook-based)] · Medium

**Q26. On a managed switch, which feature restricts a port to specific MAC addresses so that unauthorized hosts cannot connect?**

Managed switch पर कौन-सी सुविधा एक port को तय MAC addresses तक सीमित करती है ताकि अनधिकित host न जुड़ सकें?

**(A)** Port Security **(B)** VTP pruning
**(C)** Spanning Tree **(D)** Port mirroring

[Source: Practice — original (SSC CGL/IBPS exam-pattern, textbook-based)] · Medium

**Q27. Compared to a hub, a switch:**

Hub की तुलना में switch:

**(A)** Sends every frame to all ports / हर frame सब ports पर भेजता है **(B)** Sends frames only to the destination port / Frame सिर्फ़ destination port पर भेजता है
**(C)** Works only at layer 3 / सिर्फ़ layer 3 पर काम करता है **(D)** Does not need MAC addresses / MAC addresses की ज़रूरत नहीं

[Source: Practice — original (textbook-based)] · Hard

## Section C — Router / Brouter

**Q28. A router performs which of the following? (1) Packet forwarding (2) Logical addressing (3) Collision detection (4) Routing decision**

Router निम्न में से कौन-सा कार्य करता है? (1) Packet forwarding (2) Logical addressing (3) Collision detection (4) Routing decision

**(A)** 1, 2 and 4 **(B)** 2, 3 and 4
**(C)** 1 and 3 only **(D)** 1, 2, 3 and 4

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch) — Q28] · Easy

**Q29. A router drops some packets deliberately due to congestion. What is this technique called?**

Router congestion के कारण कुछ packets जानबूझकर छोड़ देता है। इस तकनीक को क्या कहते हैं?

**(A)** Delay policing **(B)** Redundancy
**(C)** Random Early Detection (RED) **(D)** Reverse flooding

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

**Q30. Which device creates a separate broadcast domain?**

कौन-सा device अलग broadcast domain बनाता है?

**(A)** Switch **(B)** Bridge
**(C)** Router **(D)** Repeater

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

**Q31. Which device operates at Layer 3 of the OSI model?**

OSI model की Layer 3 पर कौन-सा device काम करता है?

**(A)** Hub **(B)** Switch
**(C)** Router **(D)** Repeater

[Source: SSC CGL Tier-2 repeated Qs (via Oliveboard)] · Medium

**Q32. Consider: S1: The TTL field decreases at each router. S2: The MAC address changes at each hop. Which is correct?**

देखिए: S1: TTL field हर router पर घटती है। S2: MAC address हर hop पर बदलता है। कौन-सा सही है?

**(A)** Only S1 **(B)** Only S2
**(C)** Both S1 and S2 **(D)** Neither S1 nor S2

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

**Q33. A brouter is a device that can act as:**

Brouter वह उपकरण है जो बन सकता है:

**(A)** Hub + repeater **(B)** Bridge + router
**(C)** Switch + modem **(D)** Firewall + gateway

[Source: Practice — original (textbook-based)] · Medium

**Q34. A brouter forwards non-routable protocols (e.g., NetBEUI) at OSI layer _____ and routable protocols (e.g., IP) at layer 3:**

Brouter non-routable protocols (जैसे NetBEUI) को OSI layer _____ पर और routable protocols (IP) को layer 3 पर भेजता है:

**(A)** 1 **(B)** 2
**(C)** 3 **(D)** 4

[Source: Practice — original (textbook-based)] · Medium

**Q35. Which hardware forwards data packets between computer networks?**

कौन-सा hardware computer networks के बीच data packets को आगे भेजता है?

**(A)** Switch **(B)** Router
**(C)** Cable **(D)** Nibble

[Source: KVS Librarian 2023 Official Paper (via Testbook) · also CN Topic Test-03] · Hard

**Q36. Among these network devices — i. Repeater ii. Hub iii. Switch iv. Router — which are considered intelligent devices?**

इन network devices में — i. Repeater ii. Hub iii. Switch iv. Router — कौन-से intelligent devices माने जाते हैं?

**(A)** i and iv only **(B)** i and ii only
**(C)** iii and iv only **(D)** ii and iii only

[Source: Kerala PSC repeated PYQs (via pscarivukal)] · Hard

**Q37. Consider the statements about an IP router: I. A router does not modify the IP packets during forwarding. II. It is not necessary for a router to implement any routing protocol. III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the incoming fragment. Which is correct?**

IP router के बारे में कथन देखिए: I. Forwarding के दौरान router IP packets को नहीं बदलता। II. Router के लिए कोई routing protocol लागू करना आवश्यक नहीं है। III. यदि outgoing link का MTU आने वाले fragment से बड़ा हो तो router को IP fragments फिर से जोड़ने चाहिए। कौन-सा सही है?

**(A)** I and II only **(B)** I only
**(C)** II and III only **(D)** II only

[Source: Unit Test-4 Solution (OCR), p15] · Hard

**Q38. When a router forwards a packet, which header field does NOT change?**

Router के packet forward करने पर कौन-सा header field नहीं बदलता?

**(A)** TTL **(B)** Header checksum
**(C)** Source IP address **(D)** Both TTL and checksum / TTL और checksum दोनों

[Source: Practice — original (textbook-based)] · Hard

## Section D — Gateway, Firewall & NAT Devices

**Q39. Which type of firewall inspects packets at the network layer and decides based on IP addresses?**

कौन-सी firewall network layer पर packets की जाँच करके IP addresses के आधार पर निर्णय लेती है?

**(A)** Packet Filtering Firewall **(B)** Stateful Firewall
**(C)** Proxy Firewall **(D)** Application Firewall

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

**Q40. Which firewall acts as an intermediary between the client and server at the application layer?**

कौन-सी firewall application layer पर client और server के बीच मध्यस्थ (intermediary) के रूप में काम करती है?

**(A)** Packet Filtering Firewall **(B)** Stateful Firewall
**(C)** Proxy Firewall **(D)** Circuit-level Gateway

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

**Q41. If two systems use DIFFERENT protocols, which device is used to link them?**

यदि दो systems भिन्न protocols उपयोग करते हैं, तो उन्हें जोड़ने के लिए कौन-सा device उपयोग होता है?

**(A)** Repeater **(B)** Gateway
**(C)** Bridge **(D)** Hub

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Medium

**Q42. Which device can operate across all 7 layers of the OSI model?**

कौन-सा device OSI model की सभी 7 layers पर काम कर सकता है?

**(A)** Router **(B)** Hub
**(C)** Gateway **(D)** Switch

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

**Q43. Devices on one network can communicate with devices on another network via:**

एक network के devices दूसरे network के devices से किसके माध्यम से संवाद कर सकते हैं?

**(A)** File server **(B)** Utility server
**(C)** Printer server **(D)** Gateway

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Medium

**Q44. What is the role of a firewall in computer networks?**

Computer networks में firewall की भूमिका क्या है?

**(A)** Managing IP addresses **(B)** Filtering and controlling network traffic
**(C)** Resolving domain names **(D)** Assigning MAC addresses dynamically

[Source: 30 important CN MCQs (Medium blog bank)] · Medium

**Q45. What is the purpose of a proxy server in networking?**

Networking में proxy server का उद्देश्य क्या है?

**(A)** Managing IP addresses **(B)** Filtering and forwarding network traffic
**(C)** Assigning MAC addresses dynamically **(D)** Encrypting network communication

[Source: 30 important CN MCQs (Medium blog bank)] · Medium

**Q46. Which is a network security system designed to protect a trusted private network from an untrusted outside network to which it is connected?**

कौन-सी network security system trusted private network को उस जुड़े हुए untrusted outside network से बचाने के लिए बनी है?

**(A)** Bridge **(B)** Cookies
**(C)** Firewall **(D)** SSL

[Source: Random Test- Rajasthan PYQ (OCR), p6 — Q110 (options reconstructed from screenshot)] · Medium

**Q47. What kind of NAT maps one private IP to exactly one public IP?**

किस प्रकार का NAT एक private IP को ठीक एक public IP से जोड़ता है?

**(A)** Dynamic NAT **(B)** Static NAT
**(C)** PAT (Port Address Translation) **(D)** Port Forwarding

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

**Q48. A firewall that keeps track of the state of live connections is called:**

वह firewall जो चालू connections की state याद रखता है, कहलाता है:

**(A)** Packet-filter firewall **(B)** Stateful firewall
**(C)** Repeater **(D)** Modem

[Source: Practice — original (textbook-based)] · Medium

**Q49. The area where public-facing servers (web, mail) are placed for security is called:**

जहाँ public servers (web, mail) सुरक्षा हेतु रखे जाते हैं, वह क्षेत्र है:

**(A)** LAN **(B)** VLAN
**(C)** DMZ **(D)** SAN

[Source: Practice — original (textbook-based)] · Medium

**Q50. Which NAT type uses a single public IP for many internal hosts?**

कौन-सी NAT कई internal hosts के लिए एक ही public IP प्रयोग करती है?

**(A)** Static NAT **(B)** Dynamic NAT
**(C)** Overload / PAT **(D)** No NAT

[Source: Practice — original (textbook-based)] · Medium

**Q51. Port Address Translation (PAT), which maps many private IP addresses to one public IP using port numbers, is also called:**

Port Address Translation (PAT) — जो port numbers से कई private IPs को एक public IP पर map करती है — कहलाती है भी:

**(A)** NAT overload **(B)** Static NAT
**(C)** Dynamic NAT **(D)** Proxy NAT

[Source: Practice — original (textbook-based)] · Medium

**Q52. Which of the following is NOT a networking device?**

निम्न में से कौन-सा networking device नहीं है?

**(A)** Gateway **(B)** Linux
**(C)** Router **(D)** Firewall

[Source: InstrumentationTools 150 CN MCQs] · Hard

**Q53. Which of the following is NOT a networking device? (Hub / Switch / Bridge / None of these)**

निम्न में से कौन-सा networking device नहीं है? (Hub / Switch / Bridge / इनमें से कोई नहीं)

**(A)** Hub **(B)** Switch
**(C)** Bridge **(D)** None of these

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Hard

## Answer Key — sirf letters (no solutions)

**Q1:** B **Q2:** D **Q3:** A **Q4:** C **Q5:** B **Q6:** A **Q7:** A **Q8:** A **Q9:** C **Q10:** C

**Q11:** A **Q12:** A **Q13:** C **Q14:** C **Q15:** B **Q16:** A **Q17:** B **Q18:** C **Q19:** A **Q20:** C

**Q21:** B **Q22:** B **Q23:** A **Q24:** B **Q25:** A **Q26:** A **Q27:** B **Q28:** A **Q29:** C **Q30:** C

**Q31:** C **Q32:** C **Q33:** B **Q34:** B **Q35:** B **Q36:** C **Q37:** D **Q38:** C **Q39:** A **Q40:** C

**Q41:** B **Q42:** C **Q43:** D **Q44:** B **Q45:** B **Q46:** C **Q47:** B **Q48:** B **Q49:** C **Q50:** C

**Q51:** A **Q52:** B **Q53:** D